Donato E. Borrello

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Summary

Scrappy, versatile, full-stack engineer, with experience launching new products, building teams and scaling complex systems. 7 years at Google in staff and management roles, and an early hire at JW Player, scaling from 10 to over 100 employees. Successfully innovated in bootstrapped research labs, low-resource startups, and high-scale enterprise businesses.

- **Systems Architecture:** Led design of a secure and privacy-sensitive architecture for bringing Google-ID personalization to 1B+ new devices. Designed predictive ad downloading, reducing latency by 92% when real-time-bidding is enabled.
- **Startup Engineering:** Early hire at JW Player, redesigned the core data pipeline and went zero-to-one on an internal Labs product to prototype new analytics dashboards within hours.
- **High-visibility Projects:** Launched audio/video advertising for Google Assistant, Cast and GoogleTV, through design of signal collection and fraud-detection pipelines. Led implementation of GDPR/CCPA response providing enforcement and ads transparency for 10B+ ads requests/day.
- **Team Building:** Rebuilt the Video SDKs team at Google after 60% attrition. Built several teams from scratch at JW Player including an Analytics Eng team, Core Pipelines and Business Intelligence team; authored the company's career ladder and incident response plan.
- **Client-facing Roles:** Negotiated complex privacy requirements in a $1B+ revenue strategic partnership between Google and Disney. Worked directly with senior leaders and engineers at Spotify, Paramount, Bell Media, Major League Baseball, etc...

Selected Innovations

Promoted Content Selection and Delivery in a Television Platform
Utility patent (US20240314381), solving for secure user-addressability on TV platforms.

Online Sequence Alignment for Real-Time Audio Transcription by Non-Experts
White-paper on aligning multiple streams of tokenized input for optimal sequencing, published at AAAI 2012.

Real time Captioning by Groups of Non-Experts
White-paper on how partial-inputs from non-expert human workers can be combined to great results for audio transcription. Nominated for best paper at UIST.

Experience

Google

Ad Manager - Video, full-stack engineer

Staff Engineer — 2024 - Present

- Led video industry response to Privacy Sandbox for VAST advertising, by publishing a technical spec. Cited by Chromium and IAB OpenRTB working groups, and implemented by Google Ad Manager, DV360, Google Ads and (third-party) Criteo.
- Created a 'Zero-Latency' roadmap and within 6 months reduced ad playback latency by 70% on key Connected-TV segments, recognized by Paramount (CBS).
- Designed a Predictive Ads Downloading system that combines offline video playback with just-in-time ad decisioning to maximize revenue. A key innovation is predicting targeting parameters instead of the winning ad. Decreases latency by up to 92%.

- Created and led a task force to improve fraud detection on Connected-TV while improving inventory access, across 4 product areas despite teams having opposing incentives.

Tech Lead / Manager 2021 - 2024

- Lead for a suite of video developer SDKs, responsible for 10T+ ad requests/year and 3x growth in impressions in my tenure.
- Launched third-party app monetization on Google TV, through design of SDKs, rendering, signal collection and user transparency.
- Improved developer scalability by emphasizing transferrable coding patterns and automation; enabling a team of 10 to drive development across 16 binaries with minimal toil due to platform fragmentation.
- Developed the next generation of leads, including 4 promotions, while growing the team from 4 to 9 members.

Senior Software Engineer 2018 - 2021

- Designed, built, and launched to 1.5B devices a full-stack system to enable Google-ID personalization to video advertising in third-party Android apps.
- This system set the standard for user privacy and has been expanded to solve for gBid (2023), AdMob (2023), GoogleTV (2024) and Waze (2025).
- Addressed scalability concerns via multi-level caching and security concerns including nation-state attackers, spies and quantum-based crypto attacks.

JW Player
Data Analytics Platform, providing infrastructure for reporting and ML

Engineering Manager 2016 - 2018

- Interviewed, hired and built out the Data organization from 8 data engineers to 30, as part of four independent teams.
- Redesigned data pipelining architecture to reduce service outages from 0.15% to 0.01% while scaling from 15k req/sec to 60k req/sec.
- Increased release velocity from 2/month to 3/week through automation and streamlining operational overhead.
- Perform technical comparisons and break-even analysis of technology stacks, then negotiated package pricing with cloud providers including AWS, Snowflake and ModeAnalytics.

Senior Software Engineer 2014 - 2016

- Developed a high performance video player, lead for implementing VPAID2, HLS and Chromecast support.
- Redesigned the Javascript message bus for higher performance, shaving 20% off startup time, across 1 billion plays per month.
- Built a testing framework combining automated and manual tests, reducing regression testing time from 4 weeks to 1.5 weeks.

FactSet Research Systems
FinTech Engineering, primarily C++ reports

Software Engineer 2012 - 2014

- Technical Lead on several web applications, setting the record for fastest rendering time

among hundreds.

- Learned to maintain, extend, and love legacy codebases. Or at least love refactoring them.
- Created a tool *p4-compare*, which let developers compare the cyclomatic complexity of their code changes against the baseline before committing it.
- Built out a toolkit for rapidly prototyping new reports, allowing teams in New York, Norwalk and London to search and share code as reusable modules.

University of Rochester HCI Lab
ML Research on improving usefulness of human inputs

Research Scientist 2011 - 2012

- Awarded Xerox Fellowship for contributions to real-time processing of crowd inputs.
- Researched crowd-sourced real-time audio transcription for hearing-impaired internet users.
- Developed a management application for Amazon Mechanical Turk which allowed faster iteration on experiments.

Education

Bachelor of Science, Computer Science — May 2012
University of Rochester — Rochester, NY

Other Contributions

Election Reform New York 2017 - 2020
Improved democratic process for millions of NYC Voters by founding a non-profit that got Ranked Choice Voting passed for primaries in 2019.

The Nitty Gritty of Adaptive Video Transmuxing 2016
Presented to EmpireJS (400+ attendees) on optimizing video transmuxing for first frame latency.

Skills

- **Programming Languages:** `Java/Kotlin`, `C++`, `Python`, `JavaScript/TypeScript`, `SQL`
- **Leadership:** `Hiring`, `Mentoring`, `Performance Reviews`, `Technical Roadmaps`
- **Interests:** `Music`, `Civics`, `Gardening`, `Chat Bots`, `Data Visualization`
- **Favorite Places:** `New York`, `Roma`, `Nanjing`, `Turks'n Caicos`, `Tamarindo`